

Fig. 4: Pin diagram of the TCS3200 colour sensor module

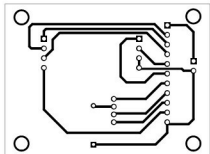


Fig. 5: Actual-size PCB layout of the RGB colour detector using TCS3200

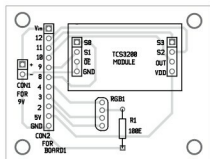


Fig. 6: Component layout of the PCB

arrays is selected separately, depending on the requirement. Hence, it is known as a programmable sensor.

The module can be used to sense a particular colour only. It contains filters for selection purpose. There is a fourth mode with no filter. With no filter, the sensor detects white light.

## Construction and testing

An actual-size, single-side PCB layout of the RGB colour detector using TCS3200 is shown in Fig. 5 and its component layout in Fig. 6.

Working of the project is simple because this is a basic circuit for interfacing a TCS3200 sensor. When red colour is kept near the sensor, it automatically detects the colour with the help of photodiode arrays and then RGB colour intensity

| TABLE I<br>TERMINAL FUNCTIONS |            |     |                                                     |
|-------------------------------|------------|-----|-----------------------------------------------------|
| Pin name                      | Pin number | I/O | Description                                         |
| GND                           | 4          |     | Power supply ground. All voltages referenced to GND |
| OE                            | 3          | I   | Enable for fo (active low)                          |
| OUT                           | 6          | O   | Output frequency (fo)                               |
| S0, S1                        | 1, 2       | I   | Output frequency scaling selection inputs           |
| S2, S3                        | 7, 8       | I   | Photodiode type selection inputs                    |
| V <sub>DD</sub>               | 5          |     | Supply voltage (5V)                                 |

| TABLE II<br>S0 AND S1 FUNCTIONS |    |                       |
|---------------------------------|----|-----------------------|
| S0                              | S1 | Output Frequency (fo) |
| L                               | L  | Power down            |
| L                               | H  | 2%                    |
| H                               | L  | 20%                   |
| H                               | H  | 100%                  |

| TABLE III<br>S2 AND S3 FUNCTIONS |    |                 |
|----------------------------------|----|-----------------|
| S2                               | S3 | Photodiode type |
| L                                | L  | Red             |
| L                                | H  | Blue            |
| H                                | L  | Clear           |
| H                                | H  | Green           |

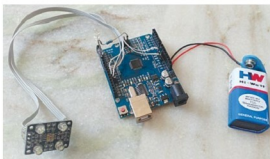


Fig. 7: Author's prototype

value is displayed in Arduino serial monitor window along with colour name. At the same time, a red LED glows in the RGB LED. Similarly, the remaining two colours (green and blue) are shown in Arduino serial monitor window and the respective colour LED glows in RGB LED.

## Software

Software is written in Arduino programming language. Arduino Uno Board1 is programmed using

Arduino IDE software. ATmega328P on Arduino Uno board comes with a pre-programmed bootloader that allows you to upload a new code to it without using an external hardware programmer.

Connect Arduino board to the PC and select the correct COM port in Arduino IDE. Compile

the program/sketch (TCS3200.ino). Select the correct board from Tools→Board menu in Arduino IDE and upload the sketch.

Load the program to the internal memory of the MCU. The sketch is at the heart of the system and carries out all major functions. It is compiled and uploaded using Arduino IDE 1.6.4.

In this project, external header files are not required for programming. It is a simple way to detect RGB colour intensities on the serial port. The author's prototype is shown in Fig. 7. **EFY**

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)



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